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A Systematic Review Protocol to Assess Artificial Intelligence Diagnostic Accuracy Performance in detecting Ischemic Stroke and Large Vessel Occlusions from Medical Imaging

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A Systematic Review Protocol to Assess Artificial Intelligence Diagnostic Accuracy Performance in detecting Ischemic Stroke and Large Vessel Occlusions from Medical Imaging

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Keywords: Artificial Intelligence; Deep Learning; Machine Learning; Cerebrovascular accident; Ischemic Stroke; Large Vessel occlusion;

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ABSTRACT

Introduction Acute ischemic stroke's timely detection and decision on optimal treatment workflow are dependent upon variable deferrals and human aptitude, bringing about mortality and dismalness. Artificial intelligence (AI) frameworks are used for automated identification and determination of strokes with accurate tissue and vessel assessment/quantification. However, the backbox AI models made the diagnosis results of AI-driven systems unexplainable and unstable. We expect to systematically review and evaluate the degree of proof to investigate the accuracy in identifying Ischemic Stroke by AI frameworks and detect Large Vessel Occlusions (LVO) in Acute ischemic stroke patients.

Methods and Analysis We will perform Systematic review and meta-analysis for the Diagnostic Accuracy Performance of AI models used to diagnose ischemic stroke and detect LVO. We will adhere to Preferred Reporting Items of Systematic reviews and Meta-analysis for Diagnostic Test Accuracy (PRISMA-DTA). Eight databases will be searched. For data screening and extraction, a modified Checklist for critical Appraisal and data extraction for systematic Reviews of prediction Modelling Studies (CHARMS), will be used by two reviewers. For included studies, Quality assessment of diagnostic accuracy studies (QUADAS-2) will be performed. A meta-analysis will be conducted if enough data are available. Summary Hierarchical receiver operating characteristic (HSROC) curves will be used to estimate summary operating points - pooled sensitivity and specificity with 95% confidence intervals (CI), if pooling is appropriate. If sufficient data is available, GRADE profiler software will be used to summarize the main findings of the systematic review is a summary of findings table.

Ethics and dissemination There is no ethical consideration with this study as the systematic review will focus on the examination of secondary information and systematic review results will be utilized to report standard procedures used in studies that assess the accuracy and completeness. We will disseminate our findings by publishing our analysis in a peer-reviewed journal and if required communicating with stakeholders of the studies and bibliographic databases.

ARTICLE SUMMARY

Strengths and limitations of this study

- Using clinical imaging published to date on diagnosing Ischemic stroke and detecting LVO, this review will be the first to systematically evaluate the performance of all AI models using diagnostic accuracy.
- The systematic review will aid to decide the potential value of utilizing AI to increase current clinical practices/methods used to diagnose Ischemic Stroke
- The systematic review will examine the results of AI models and clinicians compared, and searches to validate these findings in an external validation clinical setting.
- This decreases duplication, offers straightforwardness to the methods followed, reducing potential biases, and permits peer survey.

INTRODUCTION

Background

The increasing role of medical imaging in stroke diagnosis is gradually pushing the use of AI algorithms to extract diagnostic information automatically and aid available specialists. In clinical stroke care, Imaging modalities like advanced MRI and CT medical imaging is critical to best clinical practices. This involves excluding acute intracranial haemorrhage, identify acute ischemic stroke signs and estimate the degree of baseline ischemic damage, and detect LVO, collaterals and rule out stroke mimics, seizures, tumours, infection, and other acute neurologic diseases.

Traditional image processing techniques aid Clinicians to extract diagnostic information using classification or segmentation approaches, that generally involve shape models like atlas, active contours, region growing, etc. Region growing is a popular traditional approach to segment infarct and penumbra, which requires the practitioner to specify a seed point. Some of the limitations of traditional models are is its inability to scale its performance with more data, its customizability, etc, its strong dependence on pre-computed features, etc. These limitations are overcome by AI, as it automates the feature extraction generally done by traditional image processing techniques, for example in diagnosis, it differentiates the input images by deriving complex image features [1-3] by importing the feature semantics into the classifier needed to classify.

Several AI models are validated in ISLES challenges (2015-2018) [4-5] for Infarct and penumbra segmentation for diagnosing ischemic stroke on MRI Studies. For detection of LVO on CTA scans, [6] uses a novel AI architecture called YNet and [7] uses convolutional neural network. Also [8-9] uses traditional image processing and classifiers like random forest model to detect LVO. Computed tomography

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angiography (CTA) collateral scoring based on AI models can identify patients most likely to benefit from mechanical thrombectomy [10]. In general, Liu et al [11] reported diagnostic accuracy of AI models to be similar to health-care professionals, however, only a few studies had externally validated results with poor reporting. Murray et al [12] point out that the diagnosis of acute stroke by AI is yet to attain perfection and errors still exist. In this paper, a systematic review with Quality assessment and risk of bias was lacking as they pointed out that metrics are not generalizable between AI systems because ML systems are evaluated against different standards on different data. Given, rapid development and adoption of AI technologies in the past one year, there is a need to re-assess and perform a thorough systematic review in this area. Results of AI models that are equal or exceeding human clinicians in diagnostic performance have always generated interest but needs critical appraisal and thorough review. Concerns raised in this field pertain to the bias in study designs favouring the use of AI models, the generalizability and explainability of findings, and if the study was performed in the low-resource clinical environment, and therefore to what extent the study results apply to the real-world clinical scenarios is yet to be answered.

Objective

Upon Stoke onset, Artificial intelligence supports health specialists with necessary diagnostic information to speedup diagnosis ensuring precise intervention decisions in less time. Given that, efforts are made to implement AI models in clinical practice, we wanted to perform systematic review on the body of evidence that supports AI-based diagnosis. In this systematic review, we will provide current standards of diagnostic performance of AI models used in diagnosing ischemic stroke and detection of LVO compared with clinicians and other traditional models, looking at the issues of study design, clinical value and reporting. Our main objective is to evaluate diagnostic performance of ischemic stroke diagnosis and detection of LVO assessing potential biases. If sufficient data is available, our secondary objective will be to conduct a meta-analysis to summarize the diagnostic accuracy of AI algorithms and their comparison with clinicians and other models (deep learning vs other models), classifiers (such as CNN, SVM, random forests, LSTM, etc) in this area. If sufficient data is available, we will be calculating pooled estimates of sensitivity, and specificity and subgroup analysis and sensitivity analysis will be used.

METHODS AND ANALYSIS

Information sources

Studies will be extracted using Eight online bibliographic databases (OvidSP, Web of Science, Scopus, TRIP, ProQuest, CINAHL, IEEE, and Embase). Along with the mentioned Databases, Manipal Single Window Library Search will also be included and how to quote it will be thought off. Citation lists of included studies and relevant reviews will be checked. International Clinical Trials Registry Platform will also be searched for primary studies. As AI practices became effective from 2012, articles will be searched from January 2012 to May 2020. The reference rundown of every single qualified investigation will likewise be screened to discover any extra important sources. Also, of all included studies, the reference list will be hand searched to recognize articles that might miss by our searches using google. These articles will also be screened to discover any extra examinations (with full content) that would satisfy our standards.

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Inclusion and exclusion criteria

We will adhere to PRISMA-P [13]. Besides, reporting guidelines specific to prediction studies will be followed as appropriate [14]. Participants, interventions, comparators, and outcomes, type of studies will be included as per PICOTS [15], as shown in table 1. Letters, preprints, scientific reports, and narrative reviews in English will be included. Studies based on nonhuman samples or duplicate study data will be excluded.

PICOS	Inclusion	Exclusion
P—Population	Hyper Acute, sub-Acute Ischemic stroke patients with MR and CT images	Chronic, Trauma and other mimics
I—Intervention	Artificial Intelligence/machine learning/deep learning Algorithms	Traditional Image Processing
C—Comparison	a. Manual: Usual Health-care Professionals standard of care without AI intervention (HC vs AI) b. Semi-Automatic methods/Other Models (AI vs others)	NA
O—Outcome	**DAIGNOSE** Ischemic Stroke and **DETECT** LVO and Collaterals	Other strokes
S—Study design	Observational studies (Prospective and Retrospective cohort studies, diagnostic accuracy and case-control studies)	RCTs

Table 1 Inclusion and exclusion criteria. RCT - Randomized control trails and NA – Not Applicable.

Regularly the technique and revealing of studies assessing AI models are variable and frequently fragmented and not adhering to specific principles. In this study, we shall evaluate the Diagnostic accuracy of AI models and clinicians. We will distinguish studies presenting results with external validation or compare the performance of AI models and clinicians using the same sample and extract the reported diagnostic accuracy. If possible, contingency tables will be constructed at reporting thresholds for the diagnostic performance data. No limit will be placed on the target population but will limit our search by the disease outcome that we are interested in and the planned setting for utilizing the AI models. For the study reference standard to diagnose the absence or presence of ischemic stroke, we will use healthcare professional consensus or the standard-of-care diagnosis. The diagnosis of ischemic stroke will be assessed using Alberta stroke program early CT score (ASPECTS) or Infarct volume on CT and MR scans, as these are first in line diagnostic imaging modalities of choice in to diagnose stroke.

Search Strategy

Pertinent articles will be recognized by searching the data sources referenced for studies that are part of the inclusion criteria. The search criteria are shown in table 2. Of all included studies, the reference list will be hand searched to recognize articles that will be missed.

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Two reviewers (BK and MK) will assess the studies collected from the searches validating against the inclusion and exclusion criteria defined in Table 1. Duplicate studies will be removed.

Search Strategy used across Databases
1 = ("machine learning" OR "deep learning" OR "artificial intelligence" OR AI OR "neural network*" OR "vector machine" OR bayesian OR "deep-learning" OR "reinforcement learning" OR "reinforcement learning" OR "deep neural network" OR "deep belief network" OR "convolutional neural network" OR "recurrent neural network" OR "feedforward neural network" OR "Boltzmann machine" OR "long short term memory" OR "gated recurrent unit" OR "rectified linear unit" OR autoencoder OR backpropagation OR "multilayer perceptron" OR convnet OR "convolutional learning").af. 2 = (DICE OR "AVERAGE SYMMETRIC SURFACE DISTANCE " OR "HAUSDORFF DISTANCE" OR "symmetric surface distance " OR Jaccard OR DSC OR roc OR auc OR "goodness of fit" OR performance OR discriminate OR discrimination OR calibrate OR calibration OR accuracy OR sensitivity OR specificity OR recall OR precision OR collateral OR collaterals OR "collateral score" OR "clot burden score").ab. 3 = ("ischemic stroke" OR "large vessel occlusion" OR LVO).ab. 4 = (patient* OR subject* OR scan* OR image OR volume).ab. 5 = [2012-2020] #1 AND #2 AND #3 AND #4 AND #5

Table 2 Search terms and strategy to be used.

Data management

References from bibliographic databases will be managed by Zotero Standalone software version 5.0.88 (https://www.zotero.org/). We shall manage all the inclusion/exclusion on titles and abstracts based on predetermined eligibility criteria as mentioned in Table 1. Duplicate records will be detected and removed using the Zotero Duplication Detection feature or by using Excel macro comparing the digital object identifiers (DOI) of the searched studies manually. The articles will be identified, screened and reviewed using Zotero. Then full text articles will be retrieved into Zotero and analysed.

Study Screening and Data Extraction

Eligibility assessment assessed by screening titles and abstracts of studies retrieved from the search autonomously by two reviewers, with disagreement about inclusion/exclusion reviewed and resolved by a third reviewer (SK) by discussion. In title/ abstract review phase, initial screening will be performed by two independent reviewers (BS and MK). Based on the title and abstract information, the two reviewers will manually screen and for the finally selected articles, full text will be retrieved.
Relevant information will be extracted based on “Checklist for critical Appraisal and data extraction for systematic Reviews of prediction Modelling Studies” (CHARMS). This checklist has been adjusted according to the specific requirements of our systematic review as shown in Table 3. Two independent reviewers (BS and MK) will retrieve data using a predefined data extraction form shown in Table 3, verify the data, and resolve differences by discussion and revaluation of the article and, if the contradiction persisted, it will be referred to a third reviewer (SK) for final decision.

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1. Study Information (e.g.: Author name, Study Date, Study Source)
2. Patient eligibility (e.g.: Inclusion Criteria, Exclusion Criteria, Mean age, %LVO & %AIS participants)
3. Patient recruitment: (e.g.: number of centers, setting, phase of onset stroke, etc.)
4. Candidate predictors (e.g. Medical images from MR, CT)
5. Type of outcome (e.g., Ischemic Stroke diagnosis, LVO detection)
6. Sample size (Number of patients used in train/validation/test)
7. Model Development
    a. Model Algorithm (e.g. Neural network, Deep learning, Traditional Image processing, etc.)
8. Model Performance:
    a. Classification measures (e.g.: Sensitivity (SENS), Specificity (SPE), TP, TN, FN, FP, Threshold for SENS/SPES)
    b. Segmentation measure: (e.g.: DICE, Segmentation Accuracy, etc.)
    c. Correlation measures (e.g. MLV vs ALV, etc.)
9. Model Validation (e.g. 5-fold cross validation, Internal or external clinical validation, weather Healthcare professional comparison available)
10. Results
    a. General performance measures
    b. Any alternate AI model outcome presentations e.g.: like ASPECTS, Infarct volume, Clot burden scores or Collateral scores
11. Interpretation
    a. Models (confirmatory vs exploratory - i.e. models useful for real-world clinical practice versus models with more analysis needed)
    b. Comparison across studies discussing generalizability, strengths and limitations

Table 3 Data will be extracted from included studies using this data extraction form.

Quality Assessment

Risk of bias and disagreements will be assessed by two reviewers (BS and SK) independently, then a third reviewer (SP) will review each individual study using QUADAS-2[16]. The risk of bias will be classified as “low,” “high,” or “unclear.” The risk of bias will be summed up for each outcome across individual studies in a narrative summary in systematic review phase. The QUADAS-2 analysis will be performed in Review Manager 5.4 or STATA v16.1.

Meta-Analysis

If the studies provide adequate data and are suitable to be subjected to meta-analysis, a meta-analysis will be performed to obtain the pooled estimate of the effect. To perform metanalysis, the two reviewers (BK and MK) will extract the two-way confusion matrix table and calculated metrics like Sensitivity and Specificity (SEN, SPE) with Confidence intervals. The extracted data will be reviewed, and differences will be resolved by a third reviewer (SK or SP) by discussion. Some studies might not directly report all the required data in a 2X2 table or SEN, SPE with confidence intervals. Missing data will be calculated and filled based on the existing/reported study data, such as SEN, SPE or 2X2 confusion matrix table, and the number of cases, through the calculator in Review Manager 5.4 or Excel. Moreover, a single study might use the same AI model or clinicians for different classification tasks. Here, we will

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consider only the related tasks for this systematic review and in case of multiple comparisons, we will include a single 2X2 table reporting the highest accuracy for each one. If sufficient data available for metanalysis, the HSROC curve will be used to calculate pooled estimates - SEN and SPE with 95% CI on AI models and clinicians for diagnosis based on the 2X2 confusion matrix table. Sensitivity analysis (i.e., limiting to studies at the lowest risk of bias in a secondary analysis) will be performed to incorporate the Risk of bias assessments into synthesis whenever relevant. The HSROC curves will be plotted using STATA v16.1. Different diagnostic thresholds that will be used in the included studies may cause heterogeneity. Hence, we will consider the studies to be heterogeneous and conduct our analysis. In case we have sufficient data to determine heterogeneity, we shall use the I-squared statistic (I2 > 60% = significant heterogeneity) to assess heterogeneity, otherwise, a random-effects model will be adopted for meta-analysis. Review Manager Version 5.4 and STATA v16.1 will be used for performing a meta-analysis. If sufficient data available, we will use the GRADE approach to evaluate the certainty of evidence. Using GRADE profiler software, we will present the main findings of the systematic review is a summary of findings table.

Discussion and Conclusion

As of late, with the development of AI models, the clinicians are attempting to use them in clinical practices like the diagnosis of ischemic stroke and detection of LVO, but the accuracy metrics of these models will vary significantly. Despite several studies claiming that AI models can improve stroke diagnosis accuracy and LVO detection, the proof is constrained due to the low usage of suitable techniques and quality assessment tools providing a systematic evaluation. Especially, when performing a systematic review to evaluate AI model diagnostic performance, the review should be accurately reflecting the current performance evidence with greater scientific rigor. One significant constraint is that the detection of ischemic stroke and assessing LVO doesn't report results using different methods and outcome measures to detect or quantify the outcomes, which may lead to more optimistic results with overfitting. Moreover, the primary studies are predominantly retrospective as there are a few Randomized control trials (RCTs) in this area, and the primary studies quality will influence the systematic review evaluation quality. Exclusion of non-English studies may miss relevant articles describing AI models usage in stroke diagnosis and LVO detection in this systematic review. Overall, this is an important systematic review to assess and critically appraise evidence on AI models use in clinical practice, that shall provide impactful future direction in stroke diagnosis. The systematic review results will be published in a peer-reviewed medical journal.

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Author Contributions

Contributors SP and SK outlined the systematic review protocol and composed the manuscript. SK, BS, and MK defined the concepts, search items, data extraction process, and contributed to the final written manuscript. SP critically revised the manuscript and approved the final version.

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PRISMA-P Checklist

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	NA
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	1
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	7
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support			
Sources	#5a	Indicate sources of financial or other support for the review	8
Sponsor	#5b	Provide name for the review funder and / or sponsor	8
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	8
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	2
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	2
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	3
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	3
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	4

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Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	5
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	5
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	5
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	6
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	6
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	6
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	6
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	6
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	6
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	7

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

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In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

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Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	NA

Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	1
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	7
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support			
Sources	#5a	Indicate sources of financial or other support for the review	8
Sponsor	#5b	Provide name for the review funder and / or sponsor	8
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	8
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is	2

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		already known	
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	2
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	3
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	3
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	4
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	5
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	5
Study records - data collection	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any	5

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process		processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	6
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	6
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	6
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	6
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	6
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within	6

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		studies)	
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	7

None The PRISMA-P checklist is distributed under the terms of the Creative Commons Attribution License CC-BY 4.0. This checklist can be completed online using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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A Systematic Review Protocol to Assess Artificial Intelligence Diagnostic Accuracy Performance in detecting Acute Ischemic Stroke and Large Vessel Occlusions on CT and MR Medical Imaging

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A Systematic Review Protocol to Assess Artificial Intelligence Diagnostic Accuracy Performance in detecting Acute Ischemic Stroke and Large Vessel Occlusions on CT and MR Medical Imaging

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Keywords: Artificial Intelligence, Deep Learning, Machine Learning, Cerebrovascular accident, Ischemic Stroke, Large Vessel occlusion, Collaterals.

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ABSTRACT

Introduction Timely detection and treatment decisions are crucial to the management of acute ischaemic stroke (AIS). Any delay in diagnosis or treatment can result in a poor outcome. Artificial intelligence (AI) frameworks are used for automated identification of strokes with accurate tissue and vessel assessment/quantification. However, black-box AI models make the diagnosis results unexplainable and unstable in a clinical setting. We systematically review and evaluate the degree of proof to investigate the accuracy in diagnosing AIS and detect large vessel occlusions (LVO) by AI in AIS patients.

Methods and analysis We will perform a systematic review and meta-analysis for the diagnostic performance of AI models used to diagnose AIS and detect LVO. We will adhere to the Preferred Reporting Items of Systematic reviews and Meta-analysis for Diagnostic Test Accuracy (PRISMA-DTA). Eight databases will be searched. For data screening and extraction, a modified CHecklist for critical Appraisal and data extraction for systematic Reviews of prediction Modelling Studies (CHARMS), will be used by two reviewers. For the included studies, a Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) will be performed. A meta-analysis will be conducted if enough data are available. Hierarchical Summary Receiver Operating Characteristic (HSROC) curves will be used to estimate the summary operating points - pooled sensitivity(Sens.) and specificity(Spec.) with 95% confidence intervals (CI), if pooling is appropriate. If sufficient data is available, the Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) profiler software will be used to summarise the main findings of the systematic review as a summary of results.

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Ethics and dissemination There are no ethical considerations with this study as the systematic review focus on the examination of secondary information and systematic review results will be utilised to report on standard procedures used in studies that assess the accuracy and completeness. We will disseminate our findings by publishing our analysis in a peer-reviewed journal and if required communicating with the stakeholders of the studies and bibliographic databases.

ARTICLE SUMMARY

Strengths and limitations of this study

- This systematic review addresses a gap in the current evidence-base by providing a comprehensive assessment of the diagnostic performance of AI in AIS diagnosis and LVO detection .
- This systematic review will focus only on the use of brain imaging data based input to AI models (excluding perfusion studies).
- This systematic review will examine the results of the AI models and clinicians. It will compare them using a unique search strategy to validate these findings in both an internal and external validation clinical setting.
- This systematic review is limited to evidence from non-randomised trials and the source data used in each AI study will have an impact on the results.
- Non-English electronic databases will not be searched. This limitation may cause language bias.

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INTRODUCTION

Background

Stroke represents a major cause of global death and disability. Generally, there are two types of strokes, either ischemic (in 85% of cases) or haemorrhagic and others (in 15%). LVO variably defined as intracranial anterior and posterior circulation blockages account for 24% to 46% approximately of AIS [1] compared to other sub-types like cardioembolic, lacunar (small-vessel disease), etc. The patients with AIS, having LVO in the anterior circulation, when presented early enough, qualify for intravenous alteplase and endovascular therapy treatments.

In 2015, five international, multi-centre studies with randomised and controlled prospective trials were published in the New England Journal of Medicine [2-6]. These trails demonstrated the importance of mechanical thrombectomy (MT) compared to standard medical care in improving the prognosis of anterior circulation LVO patients. Later, the American Heart Association(AHA) /American Stroke Association (ASA) guidelines updated patient selection criteria for MT [7] based on DAWN (time window from the onset: 6-24 hours) or DEFUSE-3 (time window from the onset: 6 -16 hours) including medical image diagnosis as part of the stroke workflow (Class I, level of evidence A). The 2018 guidelines from the AHA and ASA extend the time window from 6 hours to up to 24 hours for endovascular therapy in select patients[8]. The 2019 Guidelines from AHA/ASA [9] states that patients with wake up stroke symptoms or who have an unknown time of onset > 4.5 hours from last known may benefit from IV alteplase administration within 4.5 hours of recognizing the stroke symptoms. Generally, the clinical decision is primarily based on whether ischemic stroke patients

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are sent to hospitals within the time window. “Time window” and "Tissue window" are two complementary approaches to assist decision-making[10,11]. While the tissue window is the biological timing of dynamically evolving ischemia, the time window is based on the witnessed time of stroke symptoms. The AHA/ASA guidelines recommend the use of brain imaging, for example with diffusion-weighted magnetic resonance (DW-MRI) and non-invasive CTA or MRA diagnosis of AIS and detection of LVO (Class I, level of evidence A). Hence, this study protocol predominantly focuses on AI models developed to assess tissue window on brain Imaging and imaging parameter performance to diagnose AIS and detect LVO, collaterals with internal and external validation.

In clinical stroke care, imaging modalities like advanced magnetic resonance imaging (MRI) and computed tomography (CT) medical imaging are critical in improving clinical practices. This involves excluding acute intracranial haemorrhage, identifying AIS signs and estimating the degree of baseline ischemic damage, and detecting LVO [12,13], collaterals and in the process rule out stroke mimics, seizures, tumours, infection, and other acute neurologic diseases. Hurford et al. [14] show AIS management with a CT imaging workflow.Current imaging techniques include noncontrast computed tomographyy (NCCT), CTA, computed tomographic perfusion (CTP), and MRI. Generally, DW-MRI, MRA, and perfusion MR (MRP) is used as imaging modalities in AIS. The choice of CT or MRI stroke protocol for stroke cases is hospital-specific, and no attempt was made to standardise the imaging acquisition. As per the requirement, all examinations include brain and intracranial vessel imaging, with either CTA or MRA. Additional imaging techniques, like perfusion imaging, are optional. While MRI has a greater spatial resolution to detect brain ischemia, CT imaging is widely adopted to save time in diagnosis. The increasing role of medical imaging in stroke diagnosis is gradually pushing the use of traditional image

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processing and AI algorithms to extract diagnostic information automatically and aid specialists.

Traditional image processing techniques assist clinicians in extracting diagnostic information using classification or segmentation approaches. This generally involves shape models like an atlas, active contours, region growing, etc. Region growing is a popular traditional approach to segment infarct and penumbra, which requires the practitioner to specify a seed point. Some of the limitations of traditional models are its inability to scale its performance with more data, its customizability, its strong dependence on pre-computed features, etc. These limitations are overcome by AI, as it automates the feature extraction generally done by traditional image processing techniques. For example, in diagnosis, it differentiates the input images by deriving complex image features [15-17] by importing the feature semantics into the classifier.

Several AI models are validated in ISLES challenges (2015-2018) [18,19] for infarct and penumbra segmentation for diagnosing ischemic stroke on MRI Studies. For detection of LVO on CTA scans, [16] uses a novel AI architecture called Y-Net and [20] uses a convolutional neural network(CNN). Also [21,22], uses traditional image processing and classifiers like random forest model to detect LVO. CTA collateral scoring based on AI models can select patients most likely to benefit from MT[23]. In general, Liu et al. [24] reported the AI model performance to be similar to healthcare(HC) professionals; however, only a few studies have externally validated results with poor reporting. Murray et al. [25] review the use of AI in diagnose AIS and identify LVO from January 2014 to February 2019 in three bibliographic databases - PubMed, Medline, and Embase. Though the study is based on a good inclusion/exclusion criterion, it lacks the use of formal systematic review assessment tools like assessing the risk of bias and the attempt to summarise the model

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performance via meta-analysis. In this paper, we point out that metrics are not generalisable between AI systems because ML systems are evaluated against different standards based on different data. This study also points out that the diagnosis of AIS by AI is yet to attain perfection and errors still exist. Given the rapid development and adoption of AI technologies in the past one year, there is a need to re-assess and perform a thorough systematic review in this area. The results of AI models that are equal or exceeding human clinicians in diagnostic performance have always generated interest but needs critical appraisal and thorough review. The concerns raised in this field, pertain to the bias in study designs favouring the use of AI models, the ability to simplify and explain the findings; and if the study is performed in a low-resource clinical environment. Thus, to what extent the study results apply to the real clinical scenarios is yet to be ascertained. Whereas, our protocol reviews the diagnostic performance of AI models vs clinicians in internal and external validation clinical setting and includes collateral detection on brain imaging. Suh et al. [26] review the feasibility of using various CTP to predict haemorrhagic transformation in AIS on two Ovid MEDLINE and EMBASE bibliographic databases up to 29 October 2018. The data extraction is based on study characteristics, patient characteristics and technical characteristics of the CTP like software for the CTP, the CTP parameters, and cut off values for diagnosis of haemorrhagic transformation. They concluded that high blood-brain barrier permeability and hypoperfusion status derived from the CTP are associated with haemorrhagic transformation. The meta-analysis was done on a total of 15 articles with a total of over 1134 patients. The studies were heterogeneous and Higgins I2 metrics were used to confirm the result and multiple subgroup analysis was used to work around this problem. Whereas, our protocol reviews the use of AI on brain imaging (excluding CTP) inputs to diagnose the other type of stroke - ischemic stroke and to detect LVO. Domingues et al. [27] present a general survey of the use of deep learning(DL) architectures in imaging contexts extensively covering 180

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articles from 2014 to 2019 on PET and CT modalities. They cover medical problems in the context of both 3D and 2D images and provide a good narrative summary focusing on the use of DL in medical imaging like Positron Emission Tomography (PET) and CT. The two papers examine thrombus detection in CT brain scans providing the DL model performance. The search is loosely based on a few online bibliographic databases Scopus, PubMed, Web of Science, and Springer. This study discusses interesting problems about DL models like nonavailability of image data with labels, the tendency of overfitting, the differences in imaging acquisitions, interpretability in the healthcare context, etc. The paper mentions that the studies used completely different datasets, approaches and evaluation metrics. In general, the AI models are not trained with data of the same complexity. The paper is mostly in a narrative style, lacking the use of formal systematic review assessment tools like thorough inclusion/exclusion criteria and assessing the risk of bias and meta-analysis. Whereas, our protocol reviews the use of AI on brain imaging (CT and MRI) inputs to diagnose AIS and to detect LVO and collaterals. Our protocol review will comprehensively use systematic review assessment tools based on upto-to-date systematic literature searches with assessment of the risk of bias and minimization of selection bias. Our protocol will conduct a meta-analysis to summarise the diagnostic performance of AI algorithms and their comparison with clinicians and other models if sufficient data is available.

Objective

Upon the onset of stroke, AI supports health specialists with necessary diagnostic information to speed up the diagnosis while ensuring precise intervention decisions in less time. Given that, efforts are made to implement AI models in clinical practice. We wanted to perform a systematic review of the body of evidence that supports AI-based

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diagnosis. In this systematic review, we will provide the current standards of diagnostic performance of AI models used in diagnosing AIS and detection of LVO compared with clinicians and other traditional models. It will take into account the issues of study design, clinical value and reporting. Our main objective is to evaluate the diagnostic performance of AIS diagnosis and detection of LVO assessing potential biases. If sufficient data is available, our secondary objective is to conduct a meta-analysis to summarise the diagnostic performance of AI algorithms and their comparison with clinicians and other models (DL vs other models), classifiers (such as CNN, support vector machine (SVM), random forests, long short term memory networks (LSTM), etc.). Thus, enabling us to calculate the pooled estimates of Sens. and Spec.. Also, subgroup analysis and Sens. analysis will be used in meta-analysis.

METHODS AND ANALYSIS

Information sources

The studies were extracted using eight online bibliographic databases (OvidSP, Web of Science, Scopus, TRIP, ProQuest, CINAHL, IEEE, and Embase). As AI practices became effective from 2012, articles from January 2012 to May 2020 will be searched. The reference rundown of every single qualified investigation will be screened to discover any extra important sources. Also, of all the included studies, the reference list will be manually searched to identify articles that might be missed in the google searches. These articles will also be screened to discover any extra examinations (with full content) that would satisfy our standards.

Inclusion and exclusion criteria

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We will adhere to PRISMA-P [28]. Besides, reporting guidelines specific to prediction studies will be followed as appropriate [29]. Participants, interventions, comparators, outcomes, and type of studies will be included as per PICOTS [30], as shown in Table 1. Non-human study based samples or duplicate study data will be excluded.

Table 1 Inclusion and exclusion criteria

PICOS	Inclusion	Exclusion
P—Population	AIS patients with MR and CT images	Chronic, Trauma and other mimics
I—Intervention	AI /machine learning/DL algorithms on MR and CT images	Non-image input based models
C—Comparison	a. Manual: Usual healthcare professionals standard of care without AI intervention (HC vs AI) b. Semi-Automatic methods/other models (AI vs others)	No Comparison
O—Outcome	**DIAGNOSIS** AIS and **DETECT** LVO and Collaterals	Other stroke types
S—Study design	Observational studies (prospective and retrospective cohort studies, diagnostic accuracy and case-control studies)	RCTs

RCT, Randomised Control Trails

The technique and assessment of AI based studies are variable and do not adhere to specific principles. In this study, we shall evaluate the diagnostic performance of AI

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models and clinicians on only imaging data based input. The reference standard (comparison) for identifying the outcome will be based on studies presenting results with clinician external or internal validation. Some of the studies may use other models as the reference standard. Due to a lack of fixed reference standard, we cannot compare these models, and so we will summarize AI model performance and clinician's performance separately in a systematic review and exclude from a meta-analysis.

Though diagnosing AIS outcome encompasses all outcomes, we shall perform a separate subgroup analysis to **DETECT** LVO and collaterals. This is because detection of LVO and collaterals are performed in the clinical workflow only after the initial suspicion of an AIS. The outcomes of **DIAGNOSE** AIS and **DETECT** LVO and collaterals are mapped to AI model measurements in Table 2. The above-mentioned outcome measurements are very important in the time window of < 24 hours, as these patients will be eligible for treatments like MT. The AI model performance and reference standard performance will be assessed as per the "Model Performance" defined in Table 3.

The diagnosis of AIS outcome is determined based on several independent Imaging protocols proposed like MRI/A with perfusion-weighted imaging (PWI) on everyone or NCCT +CTA + CTP for everyone or NCCT +CTA selectively, NCCT and directly go to angiography, etc. Perfusion studies (CTP/MRP) is not in the scope of this study protocol. The papers might not have AI models built for the complete imaging protocol, but still, any AI model built on any one of the imaging modalities towards the outcomes mentioned in Table 2 will be considered. However, it leads to a partial diagnosis of AIS and sub-grouped based on imaging type and the outcome measured. Similarly, in the case of detection of LVO and collaterals outcomes, AI models will be considered.

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For computing and defining infarct and penumbra volumes on CTP, there are various thresholds employed on CT and MRP images [31,32]. For example, on MR, for diffusion-weighted imaging (DWI) images to detect infarct apparent diffusion coefficient (ADC) the map is computed and the threshold is below 600x1e-6mm2/sec. Similarly, in MRP cases, the Tmax map is a threshold, below 4, 6 secs to compute penumbra. The data will be pooled accordingly based on the above differences to conduct a systematic review and meta-analysis, and if required, necessary sub-group analysis will be performed.

Detecting the LVO outcome shall be measured by a clot burden score [33,34]. Collaterals determine the outcome of all stroke. If the patient has good collaterals, it gives more time to treat. Patients selected for mechanical thrombectomy based on significant penumbra supported by collaterals with a small core can be treated up to 24 hours and beyond. Most commonly, on CT modality, the ASPECTS on collaterals (not NCCT aspects), [35] and the scores of Christoforidis et al. [36,37] and Miteff et al. [38], CTA collateral scores [39] are generally used for collateral assessment [40].

Various AI models shall be built to address these outcomes, but at a broad level, the models can be classified as Classification, Segmentation, Scoring, and Correlation models. The missing data will be calculated/derived and filled based on the existing/reported study data to perform meta-analysis as described in the meta-analysis section below.

Table 2 Outcomes mapped to imaging protocols and AI models

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Outcome	Imaging Protocols	AI Models
Diagnosis of AIS	NCCT and directly use Angiography	Scoring models – ASPECTS Segmentation models – Dense MCA sign Classification models – AIS vs non-AIS, TOAST classification, Haemorrhagic vs AIS classification
	NCCT + CTA + CTP * (optional)	NCCT – same as above CTA : Classification models – AIS vs non-AIS, same as 'Detect LVO'
	MRI/A + MRP*	MRI: Segmentation Models – Infarct volume on DWI/ADC (in cc)
Detect LVO	CTA, 4D-CTA, FD-CT, CTP*, TOF MRI, MRA	Classification models – LVO vs non-LVO Scoring Models – Clot Burden score [41]
Detect Collaterals	single phase CTA, multiphase CTA, 4D-CTA, CTP*	Scoring Models – Collateral Score [42]

MCA, Middle Cerebral Artery; 4D-CTA, 4-dimensional CT Angiogram; FD-CT, Flat Detector Computed Tomography, TOF-MRI, Time-of-flight Magnetic Resonance Imaging

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* Note: Not included in the scope of this study protocol.

The source image data used in each study will have an impact on the results causing information bias. For example, studies may use validation through resampling, internal validation by split sample followed by cross-validation or external validation on a separate dataset from a different geographical region, etc. Two reviewers (BS and MK) will assess the studies collected from the searches validating against the inclusion and exclusion criteria defined in Table 1. Studies with similar sample sizes and stroke phase will be used to assess and group the source image data. Duplicate studies will be removed.

Search Strategy

Pertinent articles will be identified by searching the data sources referenced in the studies that are part of the inclusion criteria. The search criteria are shown in Table 3. The search terms like "machine learning", "deep learning", and many more as mentioned in Table 3, are combined by Boolean operators (e.g. AND, OR) to search for title and abstract as mentioned in the query on each database. Then the results of all different search combinations were combined using AND. In Table 3, search term group #4 uses wildcard characters(*) to substitute for any other character in a string.E.g patient* finds patients, scan* finds scans. Of all included studies, the reference list will be manually searched to identify articles that might be missed.

Table 3 Search Strategy of OvidSP, Web of Science, Scopus, TRIP, ProQuest, CINAHL, IEEE, and Embase

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Search	Query
#1	"machine learning"[Title] OR "deep learning"[Title] OR "artificial intelligence"[Title] OR AI[Title] OR "neural network*"[Title] OR "vector machine"[Title] OR bayesian[Title] OR "deep-learning"[Title] OR "reinforcement learning"[Title] OR "reinforcement learning"[Title] OR "deep neural network"[Title] OR "deep belief network"[Title] OR "convolutional neural network"[Title] OR "recurrent neural network"[Title] OR "feedforward neural network"[Title] OR "Boltzmann machine"[Title] OR "long short term memory"[Title] OR "gated recurrent unit"[Title] OR "rectified linear unit"[Title] OR autoencoder[Title] OR backpropagation[Title] OR "multilayer perceptron"[Title] OR convnet[Title] OR "convolutional learning"[Title]
#2	DICE[“Abstract”] OR "AVERAGE SYMMETRIC SURFACE DISTANCE "[“Abstract”] OR "HAUSDORFF DISTANCE"[“Abstract”] OR "symmetric surface distance "[“Abstract”] OR Jaccard[“Abstract”] OR DSC[“Abstract”] OR roc[“Abstract”] OR auc[“Abstract”] OR "goodness of fit"[“Abstract”] OR performance[“Abstract”] OR discriminate[“Abstract”] OR discrimination[“Abstract”] OR calibrate[“Abstract”] OR calibration[“Abstract”] OR accuracy[“Abstract”] OR sensitivity[“Abstract”] OR specificity[“Abstract”] OR recall[“Abstract”] OR precision[“Abstract”] OR collateral[“Abstract”] OR collaterals[“Abstract”] OR "collateral score"[“Abstract”] OR "clot burden score"[“Abstract”]
#3	"ischemic stroke"[“Abstract”] OR "large vessel occlusion"[“Abstract”] OR LVO[“Abstract”]
#4	patient*[“Abstract”] OR subject*[“Abstract”] OR scan*[“Abstract”] OR image[“Abstract”] OR volume[“Abstract”]
#5	[2012-2020]

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#6	#1 AND #2 AND #3 AND #4 AND #5

Data management

References from the bibliographic databases will be managed using the Zotero Standalone software version 5.0.88 (https://www.zotero.org/). We shall manage all the inclusion/exclusion on titles and abstracts based on the predetermined eligibility criteria, as mentioned in Table 1. Duplicate records will be detected and removed using the Zotero Duplication Detection feature or by using Excel macro that will compare the digital object identifiers (DOI) of the searched studies manually. The articles will be identified, screened and reviewed using Zotero. Full-text articles will be retrieved into Zotero and analysed.

Study Screening and Data Extraction

In the title/ abstract review phase, two independent reviewers (BS and MK) will perform the initial screening manually going over the search results and select articles to retrieve the full text. Eligibility will be assessed by screening titles and abstracts of retrieved studies from the search autonomously by two reviewers. Two independent reviewers (BS and MK), will retrieve the data using a predefined data extraction form, as shown in Table 4, to verify the data and resolve the differences by discussion and revaluation of the article. Furthermore, if the contradiction persisted, it will be referred to a third reviewer (SK or SP) for the final decision. Studies with similar sample sizes and stroke phase will be used to assess and group the source image data.

The relevant information will be extracted based on CHARMS. This CHARMS checklist adjusted as per our systematic review-specific requirements, as shown in Table 4.

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Table 4 Modified CHARMS data extraction form for included studies

S.No.	Data item
1	Study Information (e. g., Author name, Study Date, Study Source)
2	Patient eligibility (e. g., Inclusion Criteria, Exclusion Criteria, Mean age, %LVO and %AIS participants)
3	Patient recruitment: (e. g., number of centres, setting, phase of onset stroke)
4	Candidate predictors (e.g., Medical images from MR, CT)
5	Type of outcome (e.g., ischemic stroke diagnosis, LVO detection)
6	Sample size (Number of patients used in train/validation/test)
7	Model Development - Model Algorithm (e.g., Neural network, DL, Traditional Image processing, etc.)
8	Model Performance: a. Classification measures (e. g., Sens., Spec., TP, TN, FN, FP, Threshold for Sens./ Spec.) b. Segmentation measure: (e. g., DICE, Accuracy, etc.) c. Scoring measures (e.g., RMSE, MSE, MAE, etc.) d. Correlation measures (e.g., MLV vs ALV, etc.)
9	Model Validation (e.g. 5-fold cross-validation, Internal or external clinical validation, weather Healthcare professional comparison available)
10	Results a. General performance measures b. Any alternate AI model outcome presentations, e. g., like ASPECTS, infarct volume, Clot burden scores or Collateral scores

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	11	Interpretation - Models (confirmatory vs exploratory - i.e. models useful for real-world clinical practice versus models with more analysis needed) Comparison across studies discussing generalizability, strengths and limitations

TP, True Positive; TN, True Negative; FN, False Negative; FP, False Positive; RMSE, Root Mean Square Error; MSE, Mean Square Error; MAE, Mean Absolute Error; MLV, Manually segmented lesion volumes; ALV, Automatically segmented lesion volumes

Table 4 Data will be extracted from included studies using this data extraction form. True Positive(TP), True Negative(TN), False Negative(FN), False Positive(FP), Root Mean Square Error (RMSE), Mean Square Error (MSE), Mean Absolute Error (MAE), Manually segmented lesion volumes (MLV), Automatically segmented lesion volumes (ALV).

Quality Assessment

The risk of bias and disagreements will be assessed by two reviewers (BS and SK) independently, then a third reviewer (SP) will review each study using QUADAS-2[43]. The risk of bias will be classified as "low," "high," or "unclear." The risk of bias will be summed up for each outcome across individual studies in a narrative summary in the systematic review phase. The QUADAS-2 analysis will be performed in Review Manager 5.4 or STATA v16.1.

Meta-Analysis

If the studies provide adequate data and are suitable to be subjected to meta-analysis, a meta-analysis will be performed to obtain the pooled estimate of the effect. To

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perform the metanalysis, the two reviewers (BK and MK) will extract the two-way confusion matrix table and calculated metrics like Sens. and Spec. with CI. The extracted data will be reviewed, and differences will be resolved by a third reviewer (SK or SP) through a discussion. Some studies might not directly report all the required data in a 2X2 table or Sens., Spec. with CI. The missing data will be calculated and filled based on the existing/reported study data, such as Sens., Spec. or 2X2 confusion matrix table, and the number of cases, through the calculator in Review Manager 5.4 or Excel. Moreover, a single study might use the same AI model or clinicians for different classification tasks. Here, we will consider only the related tasks for this systematic review. In case of multiple comparisons, we will include a single 2X2 table reporting the highest accuracy for each one. If sufficient data is available for metanalysis, the HSROC curve will be used to calculate pooled estimates - Sens. and Spec. with 95% CI on AI models and clinicians for diagnosis based on the 2X2 confusion matrix table. The Sens. analysis (i.e., limiting to studies at the lowest risk of bias in a secondary analysis) will be performed to incorporate the risk of bias assessments into synthesis whenever relevant. The HSROC curves will be plotted using STATA v16.1. Different diagnostic thresholds that will be used in the included studies may cause heterogeneity. Hence, we will consider the studies to be heterogeneous and conduct our analysis. In case we have sufficient data to determine heterogeneity, we shall use the I-squared statistic (I2 > 60% = significant heterogeneity) to assess heterogeneity; otherwise, a random-effects model will be adopted for meta-analysis. Review Manager Version 5.4 and STATA v16.1 will be used for performing a meta-analysis. If sufficient data is available, we will use the GRADE approach to evaluate the certainty of evidence. Using the GRADE profiler software, we will present the main findings of the systematic review as a summary of the findings table.

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Discussion and Conclusion

As of late, with the development of AI models, clinicians are attempting to use them in clinical practices like in the diagnosis of AIS and detection of LVO. But, the accuracy metrics of these models vary significantly. Despite several studies claiming that AI models can improve stroke diagnosis accuracy and LVO detection, the proof is constrained due to the low usage of suitable techniques and quality assessment tools, providing a systematic evaluation. Especially when performing a systematic review to evaluate AI model diagnostic performance, the review should accurately reflect the current performance evidence with greater scientific rigour.

The current study has some limitations: Firstly, the review will not include the non-diagnostic benefits of AI in stroke pathways. This includes criteria such as improving clinicians' confidence in diagnosing AIS/LVO with the help of AI clinical decision support tool and increasing the number of patients undergoing reperfusion treatments (thrombolysis or MT). Secondly, AI algorithms are dynamic, and their performance improvement is dependent on the variety and degree of data. Hence, the conclusions based on past performance may not be conclusive. Thirdly, there is a lack of RCTs or Clinical performance trails comparing AI products or various algorithms/combinations. Fourthly, there is a lack of information on source data upon which the AI algorithms are derived. This constitutes an inherent bias on the performance of AI products in clinical trials/real life. The fifth and significant constraint is that the detection of AIS and assessing LVO reports results using different methods.The sixth constraint is, though peneumbra imaging (CTP/MRP) is an important area in stroke care where AI models has been excluded in the scope of this protocol. Furthermore,study results are reported using different source data with limited public availability and diverse AI model validation approaches followed may have an impact on the quality of the systematic review. Additionally, the primary studies are predominantly retrospective

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as there are a few RCT in this area, and the primary studies quality will influence the systematic review evaluation quality. The exclusion of non-English studies may lead to an oversight of relevant articles that describe AI models usage in stroke diagnosis and LVO detection in this systematic review.

Overall, this is an important systematic review to assess and critically appraise evidence on AI models use in clinical practice. It could provide impactful insights into acute ischemic stroke diagnosis in the future. The systematic review results will be published in a peer-reviewed medical journal.

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Author Contributions

Contributors SK and SP outlined the systematic review protocol and composed the manuscript. SK, BS, and MK defined the concepts, search items, data extraction process and contributed to the final written manuscript. SP critically revised the manuscript and approved the final version.

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Reporting checklist for protocol of a systematic review.

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Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	NA

Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	1
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	21
Amendments			
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Sources	#5a	Indicate sources of financial or other support for the review	22
Sponsor	#5b	Provide name for the review funder and / or sponsor	NA
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	NA
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is	4

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		already known	
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	9
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	10
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	9
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	15
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	16
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	16
Study records - data collection	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any	17

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process		processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	17
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	17
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	18
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	6
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	18
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	18
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	18
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within	18

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		studies)	
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	19

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A Systematic Review Protocol to Assess Artificial Intelligence Diagnostic Accuracy Performance in detecting Acute Ischemic Stroke and Large Vessel Occlusions on CT and MR Medical Imaging

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A Systematic Review Protocol to Assess Artificial Intelligence Diagnostic Accuracy Performance in Detecting Acute Ischemic Stroke and Large Vessel Occlusions on CT and MR Medical Imaging

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ABSTRACT

Introduction Artificial intelligence (AI) to support Acute Ischemic stroke (AIS) diagnosis could improve patient outcomes and facilitate accurate tissue and vessel assessment. However, published evidence of AI studies report inadequately, and low explainability reduces the diagnosis results' accountability in a clinical setting. This protocol systematically reviews the accuracy of AI in AIS diagnosis and detect large vessel occlusions (LVO).

Methods and analysis We will perform a systematic review and meta-analysis for the diagnostic performance of AI models used to diagnose AIS and detect LVO. We will adhere to the Preferred Reporting Items of Systematic reviews and Meta-analysis for Diagnostic Test Accuracy (PRISMA-DTA). We will search for eight databases. For data screening and extraction, two reviewers will use a modified Checklist for Critical Appraisal and Data Extraction for Systematic Reviews of Prediction Modelling Studies (CHARMS). For the included studies, we will perform a Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) assessment. We will conduct a meta-analysis if enough data are available. We will use Hierarchical Summary Receiver Operating Characteristic (HSROC) curves to estimate the summary operating points - pooled sensitivity (Sens.) and specificity (Spec.) with 95% confidence intervals (CI) if pooling is appropriate. If sufficient data is available, we will use the Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) profiler software to summarise the main findings of the systematic review as a summary of results.

Ethics and dissemination There are no ethical considerations with this study as the systematic review focus on the examination of secondary information, and systematic

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review results will be utilized to report on standard procedures used in studies that assess the accuracy and completeness. We will disseminate our findings by publishing our analysis in a peer-reviewed journal and if required communicating with the stakeholders of the studies and bibliographic databases.

ARTICLE SUMMARY

Strengths and limitations of this study

- This systematic review addresses a gap in the current evidence-base by providing a comprehensive assessment of the diagnostic performance of AI in AIS diagnosis and LVO detection.
- This systematic review will focus only on the use of brain imaging data-based input to AI models (excluding perfusion studies).
- This systematic review will examine the results of the AI models and clinicians. It will compare them using a unique search strategy to validate these findings in both an internal and external validation clinical setting.
- This systematic review is limited to evidence from non-randomized trials, and the source data used in each AI study will impact the results.
- We will not search for Non-English electronic databases. This limitation may cause language bias.

INTRODUCTION

Background

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Stroke represents a significant cause of global death and disability. Generally, there are two types of strokes, either ischemic (in 85% of cases) or hemorrhagic and others (in 15%). Ischemic stroke with LVO variably defined as intracranial anterior and posterior circulation blockages account for 24% to 46% approximately of AIS [1]. Medical professionals should select the right treatment options for ischemic stroke patients as quickly as possible to reduce death and disability.

In 2015, five international, multi-center studies published in the New England Journal of Medicine with randomized and controlled prospective trials were published [2-6]. These trials demonstrated the importance of mechanical thrombectomy (MT) compared to standard medical care in improving the prognosis of anterior circulation LVO patients. Later, the American Heart Association(AHA) /American Stroke Association (ASA) guidelines updated patient selection criteria for MT [7] based on DAWN (time window from the onset: 6-24 hours) or DEFUSE-3 (time window from the onset: 6 -16 hours) including medical image diagnosis as part of the stroke workflow (Class I, level of evidence A). The 2018 guidelines from the AHA/ASA extend the time window from 6 hours to up to 24 hours for endovascular therapy in select patients [8]. The 2019 Guidelines from AHA/ASA [9] states that patients with wake-up stroke symptoms or who have an unknown time of onset > 4.5 hours from the last known may benefit from IV alteplase administration within 4.5 hours of recognizing the stroke symptoms. Generally, ischemic stroke patients are needed to reach treatable stroke hospitals within the time window for timely clinical treatment decisions. The time window is more and more replaced by the tissue window to assist decision-making [10,11]. While the tissue window is the biological timing of dynamically evolving ischemia, the witnessed time of stroke symptoms defines the time window. The AHA/ASA guidelines recommend using brain imaging, for example, with diffusion-weighted magnetic resonance (DW-MRI) and non-invasive CTA or MRA diagnosis of

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AIS and detection of LVO (Class I, level of evidence A). Hence, this study protocol predominantly focuses on AI models developed to assess tissue window on brain imaging and imaging parameter performance to diagnose AIS and detect LVO, collaterals with internal and external validation.

In clinical stroke care, imaging modalities like advanced magnetic resonance imaging (MRI) and computed tomography (CT) medical imaging are critical in improving clinical practices. The AIS diagnosis involves excluding acute intracranial hemorrhage, identifying AIS signs and estimating the degree of baseline ischemic damage, and detecting LVO [12,13], collaterals and in the process rule out stroke mimics, seizures, tumours, infection, and other acute neurologic diseases. Hurford et al. [14] show AIS management with a CT imaging workflow. Current imaging techniques include non-contrast computed tomography (NCCT), CTA, computed tomographic perfusion (CTP), and MRI. Also, some hospitals use DW-MRI, MRA, and perfusion MR (MRP) in AIS diagnosis. CT or MRI stroke protocol for stroke cases is hospital specific, as the image acquisition is not standardized. The increasing role of medical imaging in stroke diagnosis is gradually pushing traditional image processing and AI algorithms to automatically extract diagnostic information and aid specialists.

Several traditional image processing techniques used in AIS diagnosis include shape models like an atlas, active contours, region growing assist clinicians in extracting diagnostic information. Some of the limitations of conventional models are their inability to scale performance with more data, their customizability, strong dependence on pre-computed features. These limitations are overcome by AI, as it automates the feature extraction generally done by traditional image processing techniques. It differentiates the input images by deriving complex image features [15-17] by importing the feature semantics into the classifier. Several AI models are validated in

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ISLES challenges (2015-2018) [18,19] for infarct and penumbra segmentation for diagnosing ischemic stroke on MRI Studies. For detection of LVO on CTA scans, [16] uses a novel AI architecture called Y-Net, and [20] uses a convolutional neural network (CNN). Also [21,22], uses traditional image processing and classifiers like random forest model to detect LVO. CTA collateral scoring based on AI models can select patients most likely to benefit from MT [23].

In general, Liu et al. [24] reported the AI model performance to be similar to healthcare (HC) professionals. However, only a few studies have externally validated results with poor reporting. Murray et al. [25] review the use of AI in diagnosing AIS and identify LVO from January 2014 to February 2019 in three bibliographic databases - PubMed, Medline, and Embase. Though the study uses a loosely similar inclusion/exclusion criterion, it lacks diagnostic performance assessment of AI models with HC professionals in internal/external validation settings. Additionally, this study has not used formal systematic review assessment tools like assessing the risk of bias and summarising the model performance via meta-analysis. Suh et al. [26] review the feasibility of using various CTP to predict only hemorrhagic transformation (no ischemic changes) in AIS on two Ovid MEDLINE and EMBASE bibliographic databases up to 29 October 2018. The study performs meta-analysis on 15 articles with a total of over 1134 patients. The authors used heterogeneous articles with Higgins I2 metrics to confirm the result and used multiple subgroup analyses to solve this problem. Domingues et al. [27] present a general survey of the use of deep learning (DL) architectures in imaging contexts, extensively covering 180 articles from 2014 to 2019 on PET and CT modalities. They cover medical problems in the context of both 3D and 2D images and provide a good narrative summary focusing on the use of DL in medical imaging like Positron Emission Tomography (PET) and CT. Though the paper is mostly narrative style and lacks systematic review tools, it discusses

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interesting problems about DL models like the non-availability of image data with labels, overfitting, and the differences in imaging acquisitions, interpretability in the healthcare context.

The results of AI models that are equal or exceeding human clinicians in diagnostic performance have always generated interest but needs critical appraisal and thorough review. This systematic review aims to address the gaps identified above, like assessing the diagnostic performance of AI in AIS diagnosis and LVO detection with the use of brain imaging data-based input to AI models (excluding perfusion studies). This systematic review will examine the results of the AI models and clinicians. It will compare them using a unique search strategy to validate these findings in both an internal and external validation clinical setting. Our protocol will conduct a meta-analysis to summarise the diagnostic performance of AI algorithms and their comparison with clinicians and other models if sufficient data is available.

Objective

Upon the onset of stroke, AI supports health specialists with necessary diagnostic information to speed up the diagnosis while ensuring precise intervention decisions in less time. Given that efforts are made to implement AI models in clinical practice, we wanted to perform a systematic review of the body of evidence that supports AI-based diagnosis. In this systematic review, we will provide the current diagnostic performance standards of AI models used to diagnose AIS and detect LVO compared with clinicians and other traditional models. It will take into account the issues of study design, clinical value, and reporting. Our main objective is to evaluate the diagnostic performance of AIS diagnosis and detection of LVO, assessing potential biases. If sufficient data is available, our secondary objective is to conduct a meta-analysis to

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summarise the diagnostic performance of AI algorithms. We will compare with clinicians and other models (DL vs. other models), classifiers (such as CNN, support vector machine (SVM), random forests, long short-term memory networks (LSTM)). Thus, enabling us to calculate the pooled estimates of Sens. and Spec. Also, we will use subgroup analysis and Sens. analysis as part of our meta-analysis as required.

METHODS AND ANALYSIS

Information sources

We will extract the studies from eight online bibliographic databases (OvidSP, Web of Science, Scopus, TRIP, ProQuest, CINAHL, IEEE, and Embase). As AI practices became effective from 2012, we will search for articles from **January 2012 to May 2020**. We will screen the reference rundown of every single qualified investigation to discover any additional essential sources. We will also manually search the reference list of all the included studies to identify missed articles in the google searches. We will also screen these articles to discover any extra examinations (with full content) that would satisfy our standards.

Inclusion and exclusion criteria

We will adhere to PRISMA-P [28] and follow reporting guidelines specific to prediction studies as appropriate [29]. Patient, Interventions, Comparison, Outcome type of studies will be included as per PICOTS [30], as shown in Table 1. We will exclude non-human or duplicate study data.

Table 1 Inclusion and exclusion criteria

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PICOS	Inclusion	Exclusion
P—Population	AIS patients with MR and CT images	Chronic, Trauma and other mimics
I—Intervention	AI /machine learning/DL algorithms on MR and CT images	Non-image input-based models
C—Comparison	a. Manual: Usual healthcare professionals standard of care without AI intervention (HC vs AI) b. Semi-Automatic methods/other models (AI vs others)	No Comparison
O—Outcome	**DIAGNOSIS** AIS and **DETECT** LVO and Collaterals	Other stroke types
S—Study design	Observational studies (prospective and retrospective cohort studies, diagnostic accuracy and case-control studies)	RCTs

RCT, Randomised Control Trails

The technique and assessment of AI-based studies are variable and do not adhere to specific principles. This study shall evaluate AI models and clinicians' diagnostic performance on only imaging input. We will base the reference standard (comparison) on studies presenting results with clinician external or internal validation. Some of the studies may use other models as the reference standard. Due to a lack of a fixed reference standard, we cannot compare these models. So, we will summarize AI

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model performance and clinicians' performance separately in a systematic review and exclude it from a meta-analysis.

Though diagnosing AIS encompasses all diagnosis decisions, we shall perform a separate subgroup analysis to detect LVO and collaterals. In the clinical workflow, LVO is detected only after the initial suspicion of an AIS. Table 2 shows that the mapping of AI models to diagnosing AIS and detecting LVO and collaterals. The Measuring the diagnosis outcome mentioned in Table 1 is critical in the time window of < 24 hours, as these patients will be eligible for treatments like MT. We will assess the AI model performance and standard reference performance as per the "Model Performance" defined in Table 3.

The diagnosis of AIS is determined based on several independent Imaging protocols proposed, like MRI/A with perfusion-weighted imaging (PWI) on everyone or NCCT +CTA + CTP for everyone or NCCT +CTA selectively, NCCT and directly go to angiography. Perfusion studies (CTP/MRP) are not in the scope of this study protocol. The papers might not have AI models built for the complete imaging protocol. However, we will consider AI models built on imaging modalities towards the study outcomes mentioned in Table 2. There are various thresholds employed on CT and MRP images to compute and define infarct and penumbra volumes on CTP[31,32]. For example, on MR, for diffusion-weighted imaging (DWI) images to detect infarct apparent diffusion coefficient (ADC), the map is computed, and the threshold is below 600x1e-6mm2/sec. Similarly, in MRP cases, the Tmax map is a threshold below 4, 6 secs to compute penumbra. We will pool the data accordingly based on the above differences to conduct a systematic review and meta-analysis, and if required, we will perform the necessary sub-group analysis.

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Detection of LVO by quantification of the intracranial thrombus extent is measured using the clot burden score [33,34]. Additionally, collaterals determine the patients' outcome in all strokes. If the patient has good collaterals, it gives more time to treat. Patients selected for mechanical thrombectomy based on significant penumbra supported by collaterals with a small core can be treated up to 24 hours and beyond. Most commonly, on CT modality, the ASPECTS on collaterals (not NCCT aspects), [35] and the scores of Christoforidis et al. [36,37] and Miteff et al. [38], CTA collateral scores [39] are generally used for collateral assessment [40].

Various AI models address these study outcomes, broadly defining the problems as Classification, Segmentation, Scoring, and Correlation models. The missing data will be calculated/derived and filled based on the existing/reported study data to perform the meta-analysis described in the meta-analysis section below.

Table 2 Study Outcomes mapped to imaging protocols and AI models

Study Outcome	Imaging Protocols	AI Models
Diagnosis of AIS	NCCT and directly use Angiography	Scoring models – ASPECTS Segmentation models – Dense MCA sign Classification models – AIS vs non-AIS, TOAST classification, Haemorrhagic vs AIS classification
	NCCT + CTA + CTP * (optional)	NCCT – same as above CTA :

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		Classification models – AIS vs non-AIS, same as 'Detect LVO'
	MRI/A + MRP*	MRI: Segmentation Models – Infarct volume on DWI/ADC (in cc)
Detect LVO	CTA, 4D-CTA, FD-CT, CTP*, TOF MRI, MRA	Classification models – LVO vs non-LVO Scoring Models – Clot Burden score [41]
Detect Collaterals	single phase CTA, multiphase CTA, 4D-CTA, CTP*	Scoring Models – Collateral Score [42]

MCA, Middle Cerebral Artery; 4D-CTA, 4-dimensional CT Angiogram; FD-CT, Flat Detector Computed Tomography, TOF-MRI, Time-of-flight Magnetic Resonance Imaging

* Note: Not included in the scope of this study protocol.

The source image data used in each study will impact the results, causing information bias. For example, studies may use validation through resampling, internal validation by split sample followed by cross-validation, or external validation on a separate dataset from a different geographical region. Two reviewers (BS and MK) will assess the studies collected from the searches validating against the inclusion and exclusion criteria defined in Table 1. We will use similar sample sizes and stroke phases to assess and group the source image data.

Search Strategy

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We will identify pertinent articles by searching the data sources referenced in the studies that are part of the inclusion criteria. Table 3 shows the search criteria. The search terms like "machine learning", "deep learning", and many more, as mentioned in Table 3, are combined by Boolean operators (e.g., AND, OR) to search for title and abstract as mentioned in the query on each database. Then the results of all different search combinations were combined using AND. In Table 3, search term group #4 uses wildcard characters (*) to substitute for any other character in a string. E.g., patient* finds patients, scan* finds scans. We will manually search the reference list to identify missing articles.

Table 3 Search Strategy of OvidSP, Web of Science, Scopus, TRIP, ProQuest, CINAHL, IEEE, and Embase

Search	Query
#1	"machine learning"[Title] OR "deep learning"[Title] OR "artificial intelligence"[Title] OR AI[Title] OR "neural network*"[Title] OR "vector machine"[Title] OR bayesian[Title] OR "deep-learning"[Title] OR "reinforcement learning"[Title] OR "reinforcement learning"[Title] OR "deep neural network"[Title] OR "deep belief network"[Title] OR "convolutional neural network"[Title] OR "recurrent neural network"[Title] OR "feedforward neural network"[Title] OR "Boltzmann machine"[Title] OR "long short term memory"[Title] OR "gated recurrent unit"[Title] OR "rectified linear unit"[Title] OR autoencoder[Title] OR backpropagation[Title] OR "multilayer perceptron"[Title] OR convnet[Title] OR "convolutional learning"[Title]
#2	DICE[“Abstract”] OR "AVERAGE SYMMETRIC SURFACE DISTANCE "[“Abstract”] OR "HAUSDORFF DISTANCE"[“Abstract”] OR "symmetric surface distance "[“Abstract”] OR Jaccard[“Abstract”] OR DSC[“Abstract”]

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	OR roc[“Abstract”] OR auc[“Abstract”] OR "goodness of fit"[“Abstract”] OR performance[“Abstract”] OR discriminate[“Abstract”] OR discrimination[“Abstract”] OR calibrate[“Abstract”] OR calibration[“Abstract”] OR accuracy[“Abstract”] OR sensitivity[“Abstract”] OR specificity[“Abstract”] OR recall[“Abstract”] OR precision[“Abstract”] OR collateral[“Abstract”] OR collaterals[“Abstract”] OR "collateral score"[“Abstract”] OR "clot burden score"[“Abstract”]
#3	"ischemic stroke"[“Abstract”] OR "large vessel occlusion"[“Abstract”] OR LVO[“Abstract”]
#4	patient*[“Abstract”] OR subject*[“Abstract”] OR scan*[“Abstract”] OR image[“Abstract”] OR volume[“Abstract”]
#5	[2012-2020]
#6	#1 AND #2 AND #3 AND #4 AND #5

Data management

We will manage references from the bibliographic databases using the Zotero Standalone software version 5.0.88 (https://www.zotero.org/). We shall manage all the inclusion/exclusion on titles and abstracts based on the predetermined eligibility criteria, as mentioned in Table 1. Duplicate records will be detected and removed using the Zotero Duplication Detection feature or by using Excel macro to manually compare the digital object identifiers (DOI). The articles will be identified, screened, and reviewed using Zotero. Full-text articles will be retrieved into Zotero and analysed.

Study Screening and Data Extraction

In the title/ abstract review phase, two independent reviewers (BS and MK) will perform the initial screening, manually searching results, and select articles to retrieve the full text. Two reviewers will assess the eligibility by screening titles and abstracts of

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retrieved studies from the search autonomously. Two independent reviewers (BS and MK) will retrieve the data using a predefined data extraction form, as shown in Table 4, to verify the data and resolve the differences by discussion and revaluation of the article. Furthermore, if the contradiction persisted, it will be referred to a third reviewer (SK or SP) for the final decision. We will use similar sample sizes and the stroke phase to assess and group the source image data. We will extract the relevant information based on the adjusted CHARMS checklist adjusted as per our systematic review-specific requirements, as shown in Table 4.

Table 4 Modified CHARMS data extraction form for included studies

S.No.	Data item
1	Study Information (e. g., Author name, Study Date, Study Source)
2	Patient eligibility (e. g., Inclusion Criteria, Exclusion Criteria, Mean age, %LVO and %AIS participants)
3	Patient recruitment: (e. g., number of centres, setting, phase of onset stroke)
4	Candidate predictors (e.g., Medical images from MR, CT)
5	Type of study outcome (e.g., ischemic stroke diagnosis, LVO detection)
6	Sample size (Number of patients used in train/validation/test)
7	Model Development - Model Algorithm (e.g., Neural network, DL, Traditional Image processing, etc.)
8	Model Performance: a. Classification measures (e. g., Sens., Spec., TP, TN, FN, FP, Threshold for Sens./ Spec.) b. Segmentation measure: (e. g., DICE, Accuracy, etc.) c. Scoring measures (e.g., RMSE, MSE, MAE, etc.) d. Correlation measures (e.g., MLV vs ALV, etc.)

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	9	Model Validation (e.g., 5-fold cross-validation, Internal or external clinical validation, weather Healthcare professional comparison available)
	10	Results a. General performance measures b. Any alternate AI model presentations, e. g., like ASPECTS, infarct volume, Clot burden scores or Collateral scores
	11	Interpretation - Models (confirmatory vs exploratory - i.e., models useful for real-world clinical practice versus models with more analysis needed) Comparison across studies discussing generalizability, strengths and limitations

True Positive (TP), True Negative (TN), False Negative (FN), False Positive (FP), Root Mean Square Error (RMSE), Mean Square Error (MSE), Mean Absolute Error (MAE), Manually segmented lesion volumes (MLV), Automatically segmented lesion volumes (ALV).

Quality Assessment

The risk of bias and disagreements will be assessed by two reviewers (BS and SK) independently, then a third reviewer (SP) will review each study using QUADAS-2[43]. We will classify the risk of bias as "low," "high," or "unclear." We will sum up the risk of bias across individual studies in a narrative summary in the systematic review phase. We will perform QUADAS-2 analysis in Review Manager 5.4 or STATA v16.1.

Meta-Analysis

We will perform a meta-analysis to obtain the pooled estimate of the effect if the studies provide adequate data. The two reviewers (BK and MK) will extract the two-

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way confusion matrix table and calculated metrics like Sens. and Spec. with CI to perform the metanalysis.

A third reviewer (SK or SP) will resolve the extracted data's differences through a discussion. Some studies might not directly report all the required data in a 2X2 table or Sens., Spec. with CI. We will compute the missing data and be filled based on the existing/reported study data, such as Sens., Spec. or 2X2 confusion matrix table, and the number of cases, through the calculator in Review Manager 5.4 or Microsoft excel macros. Moreover, a single study might use the same AI model or clinicians for different classification tasks. Here, we will consider only the related tasks for this systematic review. In multiple comparisons, we will include a single 2X2 table reporting the highest accuracy for each one. If sufficient data is available for metanalysis, we will use the HSROC curve to calculate pooled estimates - Sens. and Spec. with 95% CI on AI models and clinicians for diagnosis based on the 2X2 confusion matrix table. We will perform the Sens. analysis (i.e., limiting to studies at the lowest risk of bias in a secondary analysis) will be performed to incorporate the risk of bias assessments into synthesis whenever relevant. We will use STATA v16.1 to plot HSROC curves. Different diagnostic thresholds used in the included studies may cause heterogeneity. Hence, we will consider the studies to be heterogeneous and conduct our analysis. In case we have sufficient data to determine heterogeneity, we shall use the I-squared statistic (I2 > 60% = significant heterogeneity) to assess heterogeneity. We will adopt a random-effects model for meta-analysis. We will use Review Manager Version 5.4 and STATA v16.1 to perform the meta-analysis. If sufficient data is available, we will use the GRADE approach to evaluate the certainty of evidence. Using the GRADE profiler software, we will present the systematic review's main findings as a summary of the findings table.

Patient and Public Involvement No patient involved

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DISCUCSSION AND CONCLUSION

With the development of AI models, clinicians are attempting to use AI-based software in clinical practices like diagnosing AIS and detecting LVO. But, the accuracy metrics of these models vary significantly. Despite several studies claiming that AI models can improve stroke diagnosis accuracy and LVO detection, the proof is constrained due to the low usage of suitable techniques and quality assessment tools, providing a systematic evaluation. When performing a systematic review to evaluate AI model diagnostic performance, the review should accurately reflect the current performance evidence with greater scientific rigour.

The current study has some limitations: Firstly, the review will not include the non-diagnostic benefits of AI in stroke pathways. This review excludes criteria such as improving clinicians' confidence in diagnosing AIS/LVO with the help of AI clinical decision support tools and increasing the number of patients undergoing reperfusion treatments (thrombolysis or MT). Secondly, AI algorithms are dynamic, and their performance improvement is dependent on the variety and degree of data. Hence, the conclusions based on past performance may not be conclusive. Thirdly, there is a lack of RCTs or Clinical performance trails comparing AI products or various algorithms/combinations. The primary studies are predominantly retrospective, and hence the study quality will influence the systematic review evaluation quality. Fourthly, there is a lack of information on source data used by AI algorithms that constitutes an inherent bias on the performance of AI products in clinical trials/real life. The fifth and significant constraint is that papers report AIS's detection and assessing LVO with different metrics that are not generalizable between AI systems. Additionally,

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ML systems evaluated against different standards based on different data and methods impact the systematic review quality. The sixth constraint is, exclusion of AI models on penumbra imaging (CTP/MRP), though an important area in stroke care, as the perfusion imaging is optional in AIS. The exclusion of non-English studies may lead to an oversight of relevant articles that describe AI model usage in stroke diagnosis and LVO detection in this systematic review.

Overall, this is an important systematic review to assess and critically appraise evidence on AI models use in clinical practice. It could provide impactful insights into acute ischemic stroke diagnosis in the future. We will publish the systematic review results in a peer-reviewed medical journal.

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Author Contributions

Contributors SK and SP outlined the systematic review protocol and composed the manuscript. SK, BS, and MK defined the concepts, search items, data extraction process and contributed to the final written manuscript. SP critically revised the manuscript and approved the final version.

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	1

Registration			1
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	1
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support			
Sources	#5a	Indicate sources of financial or other support for the review	21
Sponsor	#5b	Provide name for the review funder and / or sponsor	NA
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	NA
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is	4

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		already known	
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	7
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	8
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	8
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	13
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	14
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	15
Study records - data collection	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any	15

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process		processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	15
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	15
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	16
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	17
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	17
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	17
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	17
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within	17

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		studies)	
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	17

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A Systematic Review Protocol to Assess Artificial Intelligence Diagnostic Accuracy Performance in Detecting Acute Ischemic Stroke and Large Vessel Occlusions on CT and MR Medical Imaging

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A Systematic Review Protocol to Assess Artificial Intelligence Diagnostic Accuracy Performance in Detecting Acute Ischemic Stroke and Large Vessel Occlusions on CT and MR Medical Imaging

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ABSTRACT

Introduction The use of artificial intelligence (AI) to support the diagnosis of acute ischaemic stroke (AIS) could improve patient outcomes and facilitate accurate tissue and vessel assessment. However, the evidence in published AI studies is inadequate and difficult to interpret which reduces the accountability of the diagnostic results in clinical settings. This study protocol describes a rigorous systematic review of the accuracy of AI in the diagnosis of AIS and detection of large vessel occlusions (LVOs).

Methods and analysis We will perform a systematic review and meta-analysis of the performance of AI models for diagnosing AIS and detecting LVOs. We will adhere to the Preferred Reporting Items for Systematic Review and Meta-analysis Protocols (PRISMA-P) guidelines. Literature searches will be conducted in eight databases. For data screening and extraction, two reviewers will use a modified Critical Appraisal and Data Extraction for Systematic Reviews of Prediction Modelling Studies (CHARMS) checklist. We will assess the included studies using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) guidelines. We will conduct a meta-analysis if sufficient data are available. We will use Hierarchical Summary Receiver Operating Characteristic curves to estimate the summary operating points, including the pooled sensitivity and specificity, with 95% confidence intervals, if pooling is appropriate. Furthermore, if sufficient data is available, we will use Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) profiler software to summarise the main findings of the systematic review, as a summary of results.

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Ethics and dissemination There are no ethical considerations associated with this study protocol, as the systematic review focuses on the examination of secondary data. The systematic review results will be utilized to report on the accuracy, completeness, and standard procedures of the included studies. We will disseminate our findings by publishing our analysis in a peer-reviewed journal and, if required, we will communicate with the stakeholders of the studies and bibliographic databases.

ARTICLE SUMMARY

Strengths and limitations of this study

- This systematic review will address evidence gaps in the current literature by providing a comprehensive assessment of the performance of AI models in the diagnosis of AIS and LVO detection.
- This systematic review will focus only on the use of brain imaging data (excluding perfusion examinations) as input for AI models.
- This systematic review will compare the performance of AI models and clinicians, using a unique search strategy to validate the findings in both internal and external validation clinical settings.
- This systematic review will be limited to evidence from non-randomized trials, and the quality of the source data from each AI study will impact the quality of the results.
- Searches will not be conducted in non-English electronic databases, which may cause a language bias.

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INTRODUCTION

Background

Stroke is a significant cause of global death and disability. Generally, there are two types of strokes, ischaemic (accounting for 85% of cases) and haemorrhagic/other (accounting for 15% of cases). Ischaemic stroke with large vessel occlusion (LVO), variably defined as intracranial anterior and posterior circulation blockages, accounts for 24%–46% of all cases of acute ischaemic stroke (AIS).[1] Diagnosing AIS involves excluding acute intracranial haemorrhage, identifying AIS signs, estimating the degree of baseline ischaemic damage, and detecting LVOs[2,3] and collaterals. In this process, stroke mimics, seizures, tumours, infection, and other acute neurologic diseases are ruled out. Medical professionals must select the appropriate treatment for patients with ischaemic stroke as quickly as possible to reduce the risk of death and disability.

In 2015, five international, multi-centre, randomized, controlled prospective trials were published in the *New England Journal of Medicine*.[4-8] These trials demonstrated the importance of mechanical thrombectomy (MT) for improving the prognosis of patients with anterior circulation LVO compared to standard medical care. Subsequently, the American Heart Association/American Stroke Association (AHA/ASA) guidelines[9] updated the patient selection criteria for MT based on DAWN (time from onset: 6-24 hours) and DEFUSE-3 eligibility (time from onset: 6-16 hours) and included a medical image diagnosis as part of the stroke workflow (Class I; level of evidence, A).

Generally, patients with ischaemic stroke need to reach a hospital that can treat stroke within a time window to allow for urgent clinical treatment decisions. The 2018

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AHA/ASA guidelines extended the time window for endovascular therapy from 6 hours to ≤24 hours in select patients.[10] Furthermore, the 2019 AHA/ASA guidelines[11] stated that patients who wake-up with stroke symptoms >4.5 hours after they were last known to be well or have unknown time of onset may benefit from IV alteplase administration within 4.5 hours of stroke symptom recognition.

However, the time window is being increasingly replaced by the tissue window for decision-making.[12,13] While the observed onset of stroke symptoms defines the time window, the tissue window reflects the biological timing of dynamically evolving ischaemia. Thus, AHA/ASA guidelines[11] recommend using brain imaging (e.g., magnetic resonance diffusion-weighted imaging [MR-DWI], non-invasive computed tomography angiography [CTA] or magnetic resonance angiography [MRA]) in the diagnosis of AIS and detection of LVOs (Class I; level of evidence, A). Imaging modalities, such as advanced magnetic resonance imaging (MRI) and computed tomography (CT) medical imaging, have been shown to be critical in improving clinical stroke care in clinical practice. For example, Hurford et al.[14] reported on AIS management using a CT imaging workflow. Currently utilized imaging techniques include non-contrast computed tomography (NCCT), CTA, computed tomographic perfusion (CTP), and MRI. Additionally, some hospitals use MR-DWI, MRA, and perfusion MR (MRP) in the diagnosis of AIS. Notably, CT and MRI stroke protocols are hospital specific, as imaging acquisition is not standardized.

The expanding role of medical imaging in the diagnosis of stroke is gradually encouraging advancements in traditional image processing and artificial intelligence (AI) algorithms to automatically extract diagnostic information and aid specialists. Upon the onset of stroke, AI can support health specialists with the necessary diagnostic information to accelerate the diagnosis, while ensuring precise

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intervention decisions in less time. Several traditional image processing techniques used in the diagnosis of AIS include shape models, such as an atlas, active contours, and region growing, which can assist clinicians in extracting diagnostic information. However, limitations of conventional models include their inability to scale performance with more data, limited customizability, and strong dependence on pre-computed features. AI can overcome these limitations, as it automates the feature extraction process generally performed using traditional image processing techniques. It differentiates the input images, deriving complex image features by importing the feature semantics into the classifier.[15-17] Several AI models have been validated for infarct and penumbra segmentation in diagnosing ischaemic stroke on MRI in ISLES challenges (2015-2018).[18,19] For LVO detection on CTA,[16] a novel AI architecture known as Y-Net. Meijs et al.[20] uses a three-dimensional fully convolutional neural network (CNN) to segment cerebral vasculature. Other studies have used traditional image processing and classifiers such as a random-forest model to detect LVOs.[17,21] Furthermore, CTA collateral scoring based on AI models has been used to identify patients most likely to benefit from MT.[22]

Liu et al.[23] reported that AI model performance is generally similar to that of healthcare (HC) professionals. However, only a few studies have externally validated results and even these are poorly reported. Murray et al.[24] reviewed studies on the use of AI in the diagnosis of AIS and detection of LVOs published between January 2014 and February 2019 in three bibliographic databases: PubMed, Medline, and Embase. Although the review by Murray et al.[24] used loosely similar inclusion/exclusion criteria to those in our proposed study protocol, it lacked a diagnostic performance assessment of AI models compared to that of HC professionals in internal/external validation settings. Additionally, formal systematic

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review assessment tools, such as a bias risk assessment and summarization of the model performance via meta-analysis, were not used. Suh et al.[25] reviewed studies on the feasibility of using various CTP data to predict haemorrhagic transformation only (no ischaemic changes) in AIS published before 30 October 2018 in two bibliographic databases: Ovid Medline and Embase. The authors performed a meta-analysis of 15 articles, comprising over 1134 patients. The articles were heterogeneous, according to Higgins I2 metrics, and multiple sub-group analyses were performed to solve this problem. Furthermore, Domingues et al.[26] presented a general survey on the use of deep learning (DL) architectures in imaging contexts (both three-dimensional and two-dimensional), extensively covering 180 articles on positron emission tomography (PET) and CT modalities published from 2014 to 2019. Although the paper is mostly narrative in style and lacks the use of systematic review tools, it discusses interesting problems in the use of DL models in medical imaging, such as the non-availability of imaging data with labels, overfitting, differences in imaging acquisitions, and interpretability in the healthcare context.

Objective

Considerable efforts have been made to implement AI models in clinical practice, and demonstration of AI models that are equal to, or exceed, human clinicians in terms of diagnostic performance always generates interest. However, a critical appraisal and thorough review of the body of evidence supporting AI-based diagnostics are required. Therefore, in this study protocol, we propose a rigorous systematic review to address the gaps identified above. Our main objective is to provide a comprehensive assessment of the performance of AI models in the diagnosis of AIS and LVO detection using brain imaging data (excluding perfusion examinations) as input. We will compare the results of AI models and clinicians using

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a unique search strategy to validate the findings in both internal and external validation clinical settings. We will assess potential biases and take into account issues in the study design, clinical value, and reporting. Furthermore, if sufficient data are available, our secondary objective is to conduct a meta-analysis to summarise the diagnostic performance of AI algorithms in comparison to that of clinicians, other models (DL vs. other models), and classifiers (such as CNN, support vector machine [SVM], random-forest, and long short-term memory networks). Accordingly, we will be able to calculate pooled estimates of sensitivity and specificity. Additionally, we will perform sub-group analyses and a sensitivity analysis as part of our meta-analysis, as required.

METHODS AND ANALYSIS

Information sources

We will extract studies from eight online bibliographic databases: OvidSP, Web of Science, Scopus, TRIP, ProQuest, CINAHL, IEEE, and Embase. As AI methods generally became effective in 2012, we will search for articles published from January 2012 to May 2020. We will identify additional pertinent articles by searching for the data sources referenced in the studies that meet inclusion criteria. We will also manually search the reference list of all included studies to identify missed articles.

Search Strategy

Titles and abstracts will be searched using the queries indicated in Table 1. Search terms such as "machine learning" and "deep learning" will be combined by using Boolean operators (e.g., AND, OR) to create the queries. The results of different queries will be combined using the Boolean operator, AND. Wildcard characters (*)

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will be used to indicate any other character in a string (e.g., patient* includes patients, scan* includes scans; see query #4 in Table 1).

Table 1 Search strategy applied to OvidSP, Web of Science, Scopus, TRIP, ProQuest, CINAHL, IEEE, and Embase databases

Search	Query
#1	"machine learning"[Title] OR "deep learning"[Title] OR "artificial intelligence"[Title] OR AI[Title] OR "neural network*"[Title] OR "vector machine"[Title] OR bayesian[Title] OR "deep-learning"[Title] OR "reinforcement learning"[Title] OR "reinforcement learning"[Title] OR "deep neural network"[Title] OR "deep belief network"[Title] OR "convolutional neural network"[Title] OR "recurrent neural network"[Title] OR "feedforward neural network"[Title] OR "Boltzmann machine"[Title] OR "long short term memory"[Title] OR "gated recurrent unit"[Title] OR "rectified linear unit"[Title] OR autoencoder[Title] OR backpropagation[Title] OR "multilayer perceptron"[Title] OR convnet[Title] OR "convolutional learning"[Title]
#2	DICE[“Abstract”] OR "AVERAGE SYMMETRIC SURFACE DISTANCE "[“Abstract”] OR "HAUSDORFF DISTANCE"[“Abstract”] OR "symmetric surface distance "[“Abstract”] OR Jaccard[“Abstract”] OR DSC[“Abstract”] OR roc[“Abstract”] OR auc[“Abstract”] OR "goodness of fit"[“Abstract”] OR performance[“Abstract”] OR discriminate[“Abstract”] OR discrimination[“Abstract”] OR calibrate[“Abstract”] OR calibration[“Abstract”] OR accuracy[“Abstract”] OR sensitivity[“Abstract”] OR specificity[“Abstract”] OR recall[“Abstract”] OR precision[“Abstract”] OR collateral[“Abstract”] OR collaterals[“Abstract”] OR "collateral score"[“Abstract”] OR "clot burden score"[“Abstract”]
#3	"ischemic stroke"[“Abstract”] OR "large vessel occlusion"[“Abstract”] OR LVO[“Abstract”]
#4	patient*[“Abstract”] OR subject*[“Abstract”] OR scan*[“Abstract”] OR

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	image[“Abstract”] OR volume[“Abstract”]
#5	[2012-2020]
#6	#1 AND #2 AND #3 AND #4 AND #5

Inclusion and exclusion criteria

We will adhere to the Preferred Reporting Items for Systematic Review and Meta-analysis Protocols (PRISMA-P) guidelines,[27] and reporting guidelines specific to prediction studies, as appropriate.[28] In accordance with the PICOTS framework,[29] inclusion and exclusion criteria will be based on the type of patients, interventions, comparisons, and outcomes, as shown in Table 2. We will exclude data from non-human and duplicate studies.

Table 2 Inclusion and exclusion criteria

PICOS	Inclusion	Exclusion
P—Population	Patients with AIS, with MR and CT images	Stroke mimics (chronic disease, trauma, etc.)
I—Intervention	AI/machine learning/DL algorithms using MR and CT imaging data	Non-imaging-based models
C—Comparison	a. Manual: Usual HC professionals using the standard of care, without AI intervention (HC vs AI) b. Semi-Automatic methods/other models (AI vs others)	No comparisons
O—Outcome	Diagnosis of AIS and	Other stroke

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	detection of LVOs and collaterals	types
S—Study design	Observational studies (prospective and retrospective cohort studies, diagnostic accuracy studies, and case-control studies)	RCTs

AI, artificial intelligence; AIS, acute ischaemic stroke; CT, computed tomography; DL, deep learning; HC, healthcare; LVO, large vessel obstruction; MR, magnetic resonance; RCT, randomised controlled trials

Study overview

A diagnosis of AIS can be determined based on several independent imaging protocols, such as MRI/MRA with perfusion-weighted imaging, NCCT+CTA+CTP, NCCT+CTA for selected patients, or NCCT and direct angiography. Perfusion studies (e.g., CTP/MRP) are not within the scope of this study protocol. Although the identified articles might not have AI models built for all imaging protocols, we will consider AI models built on imaging modalities towards the study outcomes indicated in Table 3. Various thresholds are applied to CT and MRP images to compute and define infarct and penumbra volumes on CTP.[30,31] Furthermore, to detect infarcts on MR-DWI images, the apparent diffusion coefficient (ADC) map is computed and a threshold of $<600x1e^{-6}mm^2/sec$ is applied. Similarly, in MRP cases, a threshold of <4-6 secs is applied to the Tmax map to compute the penumbra volume. We will pool the data accordingly, based on the above differences, to conduct a systematic review and meta-analysis, and if required, we will perform the necessary sub-group analyses.

Detection of LVOs by quantification of the intracranial thrombus extent is measured using the clot burden score.[2,3] Additionally, the presence/absence of collaterals

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determines the patient's outcome for all cases of strokes. If the patient has good collaterals, there is more time to treat. Patients selected for MT based on a significant penumbra supported by collaterals, with a small core, can be treated up to 24 hours after onset and beyond. Most commonly, on CT, the Alberta Stroke Program Early CT score (ASPECTS) and collaterals (not NCCT aspects),[32] as well as the scores of Christoforidis et al.[33,34] and Miteff et al.[35], and CTA collateral scores[36] are used for collateral assessment.[37] Various AI models address these study outcomes, broadly defining the problems as classification, segmentation, scoring, and correlation models (as mapped in Table 3).

Table 3 Study outcomes mapped to imaging protocols and AI models

Study Outcome	Imaging Protocols	AI Models (type – input)
Diagnosis of AIS	NCCT and direct angiography	Scoring models – ASPECTS Segmentation models – Dense MCA sign Classification models – AIS vs non-AIS, TOAST classification, haemorrhagic vs AIS classification
	NCCT+CTA+CTP* (optional)	NCCT – same as above CTA: Classification models – AIS vs non-AIS, same as that for the 'LVO detection' outcome
	MRI/MRA + MRP*	MRI: Segmentation models – Infarct

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		volume on DWI/ADC (in cc)
LVO detection	CTA, 4D-CTA, FD-CT, CTP*, TOF-MRI, MRA	Classification models – LVO vs non-LVO Scoring models – Clot burden score[2]
Collateral detection	Single-phase CTA, multi-phase CTA, 4D-CTA, CTP*	Scoring models – Collateral score[37]

4D-CTA, 4-dimensional CT angiogram; ADC, apparent diffusion coefficient; AI, artificial intelligence; AIS, acute ischaemic stroke; ASPECTS, Alberta Stroke Program Early CT score; CT, computed tomography; CTP, computed tomographic perfusion; DWI, diffusion-weighted imaging; FD-CT, flat-detector CT, LVO, large vessel obstruction; MCA, middle cerebral artery; MRA, magnetic resonance angiography; MRI, magnetic resonance imaging; MRP, perfusion magnetic resonance; NCCT, non-contrast CT; TOF-MRI, time-of-flight MRI; TOAST, Trial of Org 10172 in Acute Stroke Treatment

* Note: Not within the scope of this study protocol.

Data management

We will manage references from the bibliographic databases using Zotero standalone software version 5.0.88 (https://www.zotero.org/). Articles will be identified, screened, and reviewed using Zotero. We will record the inclusion/exclusion status based on the predetermined eligibility criteria (Table 2). Duplicate records will be detected and removed using the Zotero Duplication Detection feature or by using an Excel macro to manually compare the digital object identifiers (DOIs). Full-text articles will be imported into Zotero and analysed.

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Study Screening and Data Extraction

In the title/abstract review phase, two independent reviewers (BS and MK) will perform the initial screening, manually searching the results, and selecting articles for full text retrieval. They will autonomously determine eligibility by screening the titles and abstracts of the retrieved studies and will retrieve data using a predefined data extraction form based on CHARMS checklist, adjusted according to our systematic review-specific requirements (Table 4). Consensus in the extracted data will be verified and any differences will be resolved by discussion and revaluation of the article. If a contradiction persists, the article will be referred to a third reviewer (SK or SP) for a final decision. The source imaging data used in each study will impact the results, potentially causing information bias. For example, studies may use validation through resampling, internal validation by split samples followed by cross-validation, or external validation using a separate dataset from a different geographical region.

We will use similar sample sizes and stroke phase to assess and group the source imaging data.

Table 4 Modified CHARMS data extraction form for the included studies

S. No.	Data item
1	Study information (e.g., author name, study date, study source)
2	Patient eligibility (e.g., inclusion criteria, exclusion criteria, mean age, %LVO, and %AIS)
3	Patient recruitment: (e.g., number of centres, setting, phase of stroke onset)
4	Candidate predictors (e.g., MR and CT medical images)

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5 Type of study outcome (e.g., ischaemic stroke diagnosis, LVO detection)

6 Sample size (number of patients used in training/validation/testing datasets)

7 Model development - model algorithm (e.g., neural network, DL, traditional image processing, etc.)

8 Model performance:

a. Classification measures (e. g., sensitivity, specificity, TP, TN, FN, FP, threshold for sensitivity/specificity)

b. Segmentation measure: (e. g., DICE, accuracy, etc.)

c. Scoring measures (e.g., RMSE, MSE, MAE, etc.)

d. Correlation measures (e.g., MLV vs ALV, etc.)

9 Model validation (e.g., 5-fold cross-validation, internal or external clinical validation, HC professional comparison available)

10 Results

a. General performance measures

b. Any alternate AI model presentations, e.g., such as ASPECTS, infarct volume, clot burden scores, or collateral scores

11 Interpretation: Models (confirmatory vs exploratory - i.e., models useful for real-world clinical practice versus models requiring more analysis); comparisons across studies discussing generalizability; strengths and limitations

AI, artificial intelligence; AIS, acute ischaemic stroke; ALV, automatically segmented lesion volumes; ASPECTS, Alberta Stroke Program Early CT score; CT, computed tomography; DL, deep learning; FN, false negative; FP, false positive; HC, healthcare; LVO, large vessel occlusions; MAE, mean absolute error; MLV, manually segmented lesion volumes; MR, magnetic resonance; MSE, mean square error; RMSE, root mean square error; TN, true negative; TP, true positive

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Quality Assessment

The risk of bias will be initially evaluated by two independent reviewers (BS and SK). A third reviewer (SP) will then review each study using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) guidelines.[38] We will classify the risk of bias as 'low', 'high', or 'unclear'. We will summarize the risk of bias across individual studies in a narrative summary in the systematic review phase. We will perform the QUADAS-2 analysis using Review Manager 5.4[39] or STATA v16.1[40] software.

Meta-Analysis

The techniques and assessments in AI-based studies vary and do not adhere to specific principles. In this study, we will evaluate the diagnostic performance of clinicians and AI models using only imaging data as input. We will base the reference standard (comparison) on studies presenting results with clinician-based external or internal validation. Some of the included studies may use other models as the reference standard. Due to a lack of a fixed reference standard, we cannot adequately compare these models. Thus, we will summarize the performance of AI models and clinicians separately in the systematic review and exclude these data from the meta-analysis.

We will perform a meta-analysis to obtain the pooled estimate of the effect if the studies provide adequate data. Two reviewers (BK and MK) will extract the two-way confusion matrix table and calculated metrics, such as the sensitivity and specificity, with confidence intervals to perform the meta-analysis. A third reviewer (SK or SP) will resolve the extracted data's differences through discussion. Some included studies might not directly report all of the required data in a 2X2 table or the

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sensitivity and specificity, with confidence intervals. We will compute the missing data using existing/reported study data, such as the sensitivity, specificity or 2X2 confusion matrix table, and the number of cases, with the calculator in Review Manager 5.4 or Microsoft excel macros. Moreover, a single study might use the same AI model or clinicians for different classification tasks. We will consider only the related tasks for this systematic review. In multiple comparisons, we will include a single 2X2 table, reporting the highest accuracy for each comparison. If sufficient data is available for a meta-analysis, we will use the Hierarchical Summary Receiver Operating Characteristic (HSROC) curve to calculate pooled estimates (sensitivity and specificity with 95% CI) for the diagnostic performance of AI models and clinicians based on the 2X2 confusion matrix table. If relevant, we will perform a sensitivity analysis (i.e., limiting the analysis to studies at the lowest risk of bias in a secondary analysis) to incorporate the risk of bias assessments into the synthesis. We will use STATA v16.1 to plot the HSROC curves. Different diagnostic thresholds used in the included studies may cause heterogeneity. Hence, we will consider the studies to be heterogeneous and will conduct our analysis accordingly. In case we have sufficient data to determine heterogeneity, we will use the I-squared statistic (I^2>60% indicates significant heterogeneity) to assess heterogeneity. We will adopt a random-effects model for the meta-analysis and use Review Manager version 5.4 and STATA v16.1 to perform the meta-analysis. If sufficient data is available, we will use the Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) approach to evaluate the certainty of the evidence. Using GRADE profiler software, we will present the main findings of the systematic review as a summary of the findings table.

Sub-group analysis

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Although the diagnosis of AIS encompasses all diagnostic decisions, we will perform a separate sub-group analysis for the detection of LVOs and collaterals. In the clinical workflow, an LVO is identified only after an initial suspicion of AIS. Table 3 shows the mapping of AI models to the diagnosis of AIS and detection of LVOs and collaterals. Obtaining a diagnosis within the time window of <24 hours is critical, as the patient is eligible for treatments such as MT. We will assess the performance of the AI model and standard reference as per the 'model performance' considerations defined in Table 4.

Patient and Public Involvement

Patients were not involved in the development of the research question, outcome measures, and study design.

ETHICS AND DISSEMINATION

There are no ethical considerations associated with this study protocol, as the systematic review focuses on the examination of secondary data. On completion of the analysis, we will prepare a manuscript for publication in a peer-reviewed medical journal.

LIMITATIONS AND STRENGTHS OF THE STUDY PROTOCOL

The study protocol has some limitations. Firstly, the review will not consider the non-diagnostic benefits of AI in stroke pathways; criteria such as improving clinicians' confidence in diagnosing AIS/LVO with the help of AI clinical decision support tools and increasing the number of patients undergoing reperfusion treatments (thrombolysis or MT) are not included. Secondly, AI algorithms are dynamic, and improvement in their performance is dependent on the variety and degree of data. Hence, AI algorithm performance assessment on few data sources may be

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inconclusive and less generalizable. Thirdly, there is a lack of random controlled trials and clinical performance trials comparing AI products or various algorithms/combinations. The primary studies are predominantly retrospective in nature and hence, the study quality will influence the quality of the systematic review. Fourthly, there is a lack of information on the source data used by AI algorithms, which constitutes an inherent bias in evaluating the performance of AI products in clinical trials/practice. Fifthly, a significant constraint is that different studies on the diagnosis of AIS and LVO detection by AI report different metrics that are not generalizable between AI systems. Additionally, AI systems are evaluated against different standards, based on different data and methods, impacting the quality of the systematic review. Sixthly, AI models on penumbra imaging (CTP/MRP) will be excluded, even though this is an important area in stroke care; however, perfusion imaging is optional in AIS. Finally, the exclusion of non-English studies may lead to an oversight of relevant articles.

With the development of AI models, clinicians are attempting to use AI-based software in clinical practices (e.g., for the diagnosis of AIS and LVO detection). However, the accuracy metrics of these models vary significantly. Despite several studies claiming that AI models can improve stroke diagnosis and LVO detection accuracy, the proof is constrained due to a low usage of suitable systematic evaluation techniques and quality assessment tools. A systematic review evaluating the diagnostic performance of AI models should accurately reflect the current performance evidence with greater scientific rigour. Overall, this systematic review is important for assessing and critically appraising the evidence on the use of AI models in clinical practice. It could provide impactful insights into the diagnosis of AIS in the future.

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Author Contributions

Contributors SK and SP outlined the systematic review protocol and composed the manuscript. SK, BS, and MK defined the concepts, search items, data extraction process and contributed to the final written manuscript. SP critically revised the manuscript and approved the final version.

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	1
Registration			1
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	1
Amendments			

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	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support			
Sources	#5a	Indicate sources of financial or other support for the review	21
Sponsor	#5b	Provide name for the review funder and / or sponsor	NA
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	NA
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	4
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	10
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	10
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	8
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	9
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	14
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	14
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for	15

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		obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	15
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	12
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	16
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	17
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	17
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	17
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	17
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	17
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	17

None The PRISMA-P checklist is distributed under the terms of the Creative Commons Attribution License CC-BY 4.0. This checklist can be completed online using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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